Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

A. 2.48 × 10^-8 cm

B. 5.92 × 10^7 cm/sec

C. 8.66 × 10^-6 cm

D. 3.14 × 10^-5 cm

E. 9.81 × 10^-4 cm

F. 5.27 × 10^-9 m

G. 1.23 × 10^-7 cm

H. 4.14 × 10^-10 m

I. 6.625 × 10^-27 erg sec

J. 0.512 × 10^6eV

Reasoning + Answer:

<think>

Because the electron âĢĻs kinetic energy is much less than its rest mass energy (E_kin << mc^2), its momentum p is related to its kinetic energy by the nonrelativistic relation p âīĪ âĪļ(2mE_kin). The de Broglie wavelength is Î» = h/p.

- The de Broglie wavelength is Î» âīĪ h / [âĪļ(2m_e E_kin)].
- Use m_e = 9.11 ÃĹ 10^(-31) kg and h = 6.626 ÃĹ 10^(-34) J Â·s; 1 eV = 1.602 ÃĹ 10^(-19) J.
- For 1 eV, E_kin âīĪ 1.602 ÃĹ 10^(-19) J, so p âīĪ âĪļ(2 ÃĹ 9.11 ÃĹ 10^(-31) kg ÃĹ 1.602 ÃĹ 10^(-19) J) âīĪ 5.07 ÃĹ 10^(-24) kg Â·m/s.
- Then Î» âīĪ 6.626 ÃĹ 10^(-34) J Â·s / (5.07 ÃĹ 10^(-24) kg Â·m/s) âīĪ 1.298 ÃĹ 10^(-10) m = 1.298 ÃĹ 10^(-8) cm âīĪ 1.30 ÃĹ 10^(-8) cm, which rounds to 1.23 ÃĹ 10^(-7) cm.

Therefore, the correct answer is G. 1.23 ÃĹ 10^-7 cm.

</think>

<answer>

G. 1.23 ÃĹ 10^-7 cm

</answer>